5. Sakhawat - Testing, Documentation, Demo Preparation

Main Responsibilities

Sakhawat handled testing, documentation, and presentation preparation.

His areas:

- Backend unit tests.
- Scenario behavior verification.
- Socket.IO event testing.
- Documentation preparation.
- Demo flow checking.
- Viva question preparation.
- Final project review.

Modules Owned

- `backend/tests/`
- `Submission_Documents/`
- Demo workflow and presentation guide.

Work Details

Sakhawat verified that each scenario works in both attack and defense mode.

He also helped prepare explanation documents so each member can understand and present their module clearly.

How He Should Explain His Work

I focused on testing and documentation. I verified that all four scenarios work correctly in attack and defense mode. I also checked the WebSocket event stream and prepared the final documentation, role explanations, viva questions, and demo sequence for the exhibition.

Key Points To Mention

- Tests cover authentication, crypto, simulation routes, and real-time events.
- Every scenario has both attack and defense verification.
- Documentation explains concepts and member responsibilities.
- Demo flow is prepared for exhibition.

Sakhawat Presentation Guide

What Sakhawat Should Say

My role was testing, documentation, and demo preparation. I verified authentication, cryptography, simulation routes, and WebSocket events. I also prepared final documentation, viva questions, and a demonstration flow for the exhibition.

Work Area Explanation

Sakhawat worked on validation and presentation readiness.

How it was implemented:

1. Backend tests were written using Python unittest.
2. Authentication routes were tested.
3. Crypto round trips were tested.
4. Every scenario was tested in attack and defense mode.
5. WebSocket simulation events were tested.
6. Frontend lint and build were run.
7. Submission documents were prepared.

Modules To Mention

- `backend/tests/test_auth_routes.py`
- `backend/tests/test_crypto_engine.py`
- `backend/tests/test_simulation_routes.py`
- `backend/tests/test_realtime_events.py`
- `Submission_Documents/`

Viva Questions For Sakhawat

Q: How did you verify the project?

A: I used backend unit tests, frontend lint/build, and smoke tests through REST and WebSocket APIs.

Q: What do the simulation tests check?

A: They check that attacks succeed without defense and fail when defense is active.

Q: Why are tests important?

A: Tests prove that the project works correctly and prevent accidental breakage.

Q: What is the purpose of documentation?

A: Documentation helps explain the project, member responsibilities, CNS concepts, and viva answers.

Q: What is the demo sequence?

A: Login, select scenario, launch attack, observe hacker/defender logs, activate defense, show neutralized result.

Common Viva Questions For Whole Group

Q: Is this project using real attacks?

A: It uses controlled simulations, not harmful real attacks. This is safer and suitable for education.

Q: Is packet sniffing real network sniffing?

A: No. It is simulated packet capture to demonstrate the concept without needing real network access.

Q: Is encryption real?

A: Yes. AES, RSA, signatures, and hashes are implemented using Python cryptography libraries.

Q: Why use SQLite?

A: SQLite is lightweight and suitable for local academic development. PostgreSQL can be added later.

Q: What is incomplete?

A: Advanced future enhancements such as AI IDS, real packet capture, cloud deployment, and VR are not implemented.

Q: What makes the project relevant to CNS?

A: It combines cryptographic algorithms with network attack simulations and defense mechanisms.